

Coding Questions

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0.0.1 1. How can you create a Numpy array? What is the shape of a Numpy array and how can you get it?

We can create the numpy array by using `numpy.array(The list name)` , the shape of the numpy array specify the no. of rows and columns of the array , We can get it by `numpy.shape(the array name)`

0.0.2 2. How can you perform element-wise operations on Numpy arrays (list the possible element-wise operations) ?

you can use it like you use it with the numbers ex: `**` used as power and `+` used for adding and so on.or there is specific functions for it like: `np.add()`,`np.subtract()`,`np.multiply()`,`np.divide()`,`np.power()`

0.0.3 3. How to sum all elements of one array ?

We can sum the elements of the array by using `numpy.sum(the_name_of_the_array)`

0.0.4 4. How to transpose a Numpy array?

We can transpose the elements of the array by using `numpy.transpose(the_name_of_the_array)`

0.0.5 5. What transpose do to the Numpy array?

flips a matrix over its diagonal; that is, it switches the row and column indices of the matrix A by producing another matrix

0.0.6 6. Having matrixA and matrixB both with shape (3,2) how to create a numpy array withshape (2,3,2) using np.concatenate?

```
[1]: import numpy as np
matrixA= np.random.randint(100, size=(3,2))
matrixB= np.random.randint(100, size=(3,2))
matrixC=np.concatenate((matrixA, matrixB), axis=0, out=None, dtype=None,
    ↪casting="same_kind")
print("The new matrix will be:\n",matrixC)
```

The new matrix will be:

```
[[22 82]
 [32 73]
 [94  8]
 [41 29]
 [46 88]
 [62 87]]
```

0.0.7 7. What is the difference between reshape and resize in Numpy?

reshape() doesn't modify the data of the numpy array, but resize() changes it.

0.0.8 8. How can you extract the diagonal elements of a Numpy array?

by using numpy.diag(Name_of_the_array)

0.0.9 9. What is the difference between np.dot and np.matmul in Numpy?

1-numpy.matmul() is Matrix product of two arrays, numpy.dot () is a dot product of two arrays. The two arrays must be of same size and the output is the same size

0.0.10 10. How can you perform element-wise comparison of two Numpy arrays (list the possible element-wise comparisons)?

np.fmin(arr1, arr2) ,numpy.logical_and(arr1, arr2) ,numpy.logical_or(arr1, arr2)
 ,numpy.greater(arr1,arr2) (Return the truth value of (x1 > x2)) ,numpy.less(arr1,arr2) (Return the truth value of (x1 < x2))